RED PAPER

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SUBJECT: PWR-SMR-2026-01 Dataset Specifications
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1.0 Safety, Scope, and Use Restrictions

The PWR-SMR-2026-01 dataset is provided strictly for research, model development, design-space exploration, benchmarking, and offline validation workflows. It is not intended for plant operations, real-time decision support, safety-critical automation, licensing analysis, or operator guidance.

This dataset consists of high-fidelity Monte Carlo neutron transport results generated for static reactor states. The published cases represent offline simulation outputs and do not constitute a reactor control model, a kinetics model, or a validated basis for transient safety decisions.

1.1 Authorized Uses

- Training and evaluating machine-learning models for classification, detection, or surrogate modeling.
- Benchmarking numerical methods, data pipelines, and scientific computing workflows.
- Supporting non-operational reactor physics research, education, and design exploration.
- Reproducing published dataset results within controlled R&D environments.

1.2 Prohibited Uses

- Direct or indirect integration into reactor control systems, protection systems, PLC logic, or live plant monitoring.
- Use in operational setpoint generation, autonomous actuation, or closed-loop control.
- Use as a substitute for licensed safety analysis, plant procedures, or engineering sign-off.
- Deployment in any environment where model error, data corruption, or misuse could affect physical plant behavior.

1.3 Technical Limitation Notice

The dataset represents static, time-independent Monte Carlo k-eigenvalue solutions. It does not model reactor kinetics, delayed neutron precursor dynamics, or time-dependent transient behavior required for real-time safety analysis. Absence of plant instrumentation context, transient modeling, and site-specific licensing assumptions limits applicability to offline research use only.

1.4 User Responsibility

Users are responsible for independent verification, checksum validation, and fitness-for-purpose review before any downstream use. Accessing or using the dataset constitutes acceptance of these restrictions.

2.0 Reactor Physics Specifications

2.1 Core Architecture

The target geometry models a generic small modular reactor (SMR) using standard pressurized water reactor (PWR) physics principles and industry-standard dimensions.

- **Moderator:** Light water (H_2O) at 15.5 MPa
- **Spectrum:** Thermal neutron spectrum
- **Cycle:** Fresh fuel (beginning of life – BOL)

2.2 Lattice Configuration

The core is constructed using a standard 17×17 PWR fuel assembly lattice.

- **Fuel rods:** 264 rods per assembly
- **Guide tubes:** 24 locations (Zircaloy-4)
- **Instrument tube:** 1 central location
- **Pitch:** 1.26 cm pin pitch (standard PWR)

2.3 Material Composition

- **Fuel:** Uranium dioxide (UO_2) ceramic pellets
- **Enrichment:** <5.00 wt% U-235 (LEU – low enriched uranium, civilian grade)
- **Cladding:** Zircaloy-4 (Zr-Sn-Fe-Cr alloy)
- **Coolant:** Borated water (variable boron concentration for criticality control)

2.4 Simulation Engine

All data points are generated using **OpenMC**, a state-of-the-art, open-source Monte Carlo neutron transport code. Unlike deterministic codes (diffusion theory), OpenMC tracks individual neutron histories in continuous energy, preserving the stochastic nature of fission.

- **Cross-section library:** ENDF/B-VIII.0

2.5 Transport Fidelity

Every simulation in this dataset (1,518 cases) was run with high-performance parameters to ensure statistical convergence:

- **Particles per batch:** 50,000
- **Total batches:** 250
- **Active batches:** 200 (first 50 discarded for source convergence)
- **Total histories:** 12,500,000 neutron histories per simulation

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3.0 Verification and Validation Summary

The PWR-SMR-2026-01 dataset was verified and validated for offline reactor-physics research and AI/ML model-development use within the scope of static Monte Carlo state calculations. Verification activities addressed source convergence, batch correlation, and uncertainty characterization. Validation activities addressed expected physics trends and comparison against a representative fresh-fuel reference case.

3.1 Numerical Verification

Source Convergence: Active-batch Shannon entropy drift measured at 0.0419%, well below the acceptance threshold of 0.1%. Source distribution converged to the fundamental eigenmode before tallying began.

Statistical Uncertainty: Active batches exhibited lag-1 autocorrelation of 0.62. Nominal uncertainty of ±27 pcm was conservatively corrected to ±56 pcm to account for reduced effective sample size (N_{eff} ≈ 47 from 200 active batches). This corrected uncertainty should be used when assessing anomaly separability and model-detection limits.

3.2 Physics Validation

Physics validation confirmed the expected negative reactivity response for fuel temperature increase, control-rod insertion, and coolant void increase in the modeled regime:

- **Fuel temperature (Doppler feedback):** Increasing fuel temperature results in decreased reactivity due to Doppler broadening of U-238 resonance capture. Measured coefficient: -0.081 pcm/K.
- **Control rod worth:** Insertion of control rods produces significant negative reactivity. Measured differential rod worth: -3.91 pcm/cm.
- **Coolant void coefficient:** Reducing coolant density decreases reactivity, consistent with an under-moderated PWR regime. Measured coefficient: -0.055 pcm/%void.

3.3 Benchmark Validation

Benchmark comparison of the nominal case yielded a bias of -339.7 pcm relative to the selected fresh-fuel reference (k_{eff} = 1.27000), which lies within the stated acceptance band of ±1000 pcm.

Result: The dataset demonstrates numerical stability, correct physics trends, and benchmark fidelity suitable for offline research, benchmarking, and model-development applications within the documented scope.

4.0 Anomaly Detection Regimes

4.1 Mixed-Detectability Framework

The dataset operates in two distinct detectability regimes based on the separability index $Z = |\Delta\rho|/\sigma_{corrected}$, where Δρ is the reactivity defect and $\sigma_{corrected} = \pm\ 56$ pcm.

4.2 Regime A: Macro-Anomalies

- **Examples:** Major control rod drops (≥50 cm), severe voiding (>20%)
- **Separability:** Z > 3.0
- **Detection approach:** Single-frame models (CNNs, MLPs) can detect these anomalies in individual snapshots due to strong signal-to-noise ratio.

4.3 Regime B: Micro-Perturbations

- **Examples:** Temperature drifts (+200 K), minor voiding (<20%), slight rod misalignments (<50 cm)
- **Separability:** Z < 3.0
- **Detection challenge:** Signal is partially or fully masked by the statistical noise floor.
- **Detection approach:** Requires temporal models (LSTMs, Transformers), trend-based analysis, denoising autoencoders, or aggregation across multiple samples to extract signal from noise.

4.4 Application Guidance

Users should select model architecture based on target anomaly class. Single-frame perceptrons are not recommended for micro-perturbation detection without preprocessing or sequence integration.

5.0 Data Integrity and Distribution

5.1 Cryptographic Verification

Every file in this dataset (1,518 simulations × 4 files/simulation) is cryptographically signed using SHA-256 (256-bit Secure Hash Algorithm). A master manifest file is provided in the root directory.

Verification protocol:

1. Users must verify file integrity against the manifest before use.
2. If hash verification fails, the file is compromised and must be deleted immediately.
3. Do not use unverified files for training or analysis.

5.2 File Architecture

Open, non-proprietary standards are used to ensure long-term reproducibility:

- **Data container:** HDF5 (.h5) – high-performance hierarchical data format, natively supported by Python (h5py) and Pandas
- **Configuration:** XML (.xml) – standard OpenMC geometry and material definitions
- **Metadata:** JSON (.json) – human-readable simulation logs and V&V tags

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6.0 Licensing and Access Tiers

6.1 Open-Access Framework (PWR-SMR-2026-01) To accelerate global nuclear AI research and enable reproducible benchmarking, KEFF Data provides the complete PWR-SMR-2026-01 dataset (all 1,518 simulation cases, including raw HDF5 data, Shannon entropy meshes, and full XML configuration inputs) entirely open-access for non-commercial applications.

- **Academic & Research Tier**
    - **Access:** Full 1,518 simulations (Data + Inputs + Manifests).
    - **License:** Creative Commons Attribution-NonCommercial 4.0 International (**CC BY-NC 4.0**).
    - **Cost:** $0 (Free for academic, educational, and non-commercial R&D workflows).

6.2 Commercial Use & Licensing Inquiries Any commercial entity, corporate R&D division, or national laboratory wishing to utilize this dataset, its underlying methodology, or its outputs for proprietary development, commercial product training, or operational benchmarking must obtain explicit written authorization and a commercial license from KEFF Data.

- **Commercial Evaluation Tier**
    - **Access:** Licensing rights for commercial exploitation of the PWR-SMR-2026-01 framework and custom data pipeline support.
    - **License:** Proprietary Commercial License.
    - **Cost:** Contact contact@keffdata.com for corporate licensing structures.

7.0 Known Limitations

1. **Statistical noise floor:** The corrected uncertainty of ±56 pcm establishes the single-frame detection threshold. Perturbations below this magnitude require trend analysis or temporal integration for reliable detection.
2. **Batch autocorrelation:** Active batches exhibit suboptimal independence ($\rho = 0.62$). While mean values are accurate, uncertainty quantification has been conservatively adjusted.
3. **Static modeling:** Dataset represents k-eigenvalue solutions only. It does not include reactor kinetics, transient behavior, or delayed neutron precursor modeling.
4. **Fresh fuel only:** All cases represent beginning-of-life conditions. Burnup effects are not included.

8.0 Publication and Citation

This dataset is published by KEFF Data for the purpose of advancing nuclear AI research and enabling reproducible benchmarking of anomaly detection methods.

Recommended citation:

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SIGNED & VERIFIED

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